

## Score-based diffusion: Forward SDE and Reverse ODE

Let  $(x_t)_{t \in [0, T]}$  be a diffusion process on  $\mathbb{R}^d$  driven by a standard Wiener process  $(w_t)$ .

**Theorem 0.1** (Forward SDE and its marginal evolution): Assume the **forward** diffusion is defined by the Itô SDE

$$dx_t = f(x_t, t) dt + g(t) dw_t, \quad t \in [0, T],$$

where  $w_t$  is a standard Wiener process in  $\mathbb{R}^d$ . The initial distribution is  $x_0 \sim p_0$  (the data distribution), drift  $f : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ , and scalar diffusion coefficient  $g : [0, T] \rightarrow (0, \infty)$ .

Then the marginal density  $p_t(x)$  of  $x_t$  exists and satisfies the **Fokker–Planck (forward Kolmogorov) PDE**:

$$\partial_t p_t(x) = -\nabla \cdot (f(x, t) p_t(x)) + \frac{1}{2} g(t)^2 \Delta p_t(x), \quad \forall x \in \mathbb{R}^d.$$

**Lemma 0.1** (Itô's Lemma for test functions): Let  $x_t$  be the process defined above and  $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$  be a smooth function ( $C^2$ ). The stochastic differential of  $\varphi(x_t)$  is given by:

$$d\varphi(x_t) = \left[ \nabla \varphi(x_t) \cdot f(x_t, t) + \frac{1}{2} g(t)^2 \Delta \varphi(x_t) \right] dt + g(t) \nabla \varphi(x_t) \cdot dw_t.$$

*Proof:* Let  $\varphi(x)$  be an arbitrary smooth test function with compact support ( $\varphi \in C_c^\infty(\mathbb{R}^d)$ ).

We define the expectation  $E(t) := \mathbb{E}[\varphi(x_t)]$ . Integrating the SDE form given in Lemma 0.1 from 0 to  $t$  and taking the expectation:

$$\begin{aligned} \mathbb{E}[\varphi(x_t)] - \mathbb{E}[\varphi(x_0)] &= \mathbb{E} \left[ \int_0^t \left[ f(x_s, s) \cdot \nabla \varphi(x_s) + \frac{1}{2} g(s)^2 \Delta \varphi(x_s) \right] ds \right] \\ &\quad + \mathbb{E} \left[ \int_0^t g(s) \nabla \varphi(x_s) \cdot dw_s \right]. \end{aligned}$$

The second term (the stochastic integral) vanishes because the Itô integral with respect to a Wiener process is a martingale (and has zero expectation starting from 0).

Differentiating the remaining term with respect to  $t$ :

$$\frac{d}{dt} \mathbb{E}[\varphi(x_t)] = \mathbb{E} \left[ f(x_t, t) \cdot \nabla \varphi(x_t) + \frac{1}{2} g(t)^2 \Delta \varphi(x_t) \right].$$

Now, we rewrite the expectation using the probability density  $p_{t(x)}$ :

$$\int_{\mathbb{R}^d} \varphi(x) \partial_t p_{t(x)} dx = \int_{\mathbb{R}^d} \left[ f(x, t) \cdot \nabla \varphi(x) + \frac{1}{2} g(t)^2 \Delta \varphi(x) \right] p_{t(x)} dx.$$

We apply integration by parts to move the derivatives from the test function  $\varphi$  to the density  $p_t$ . Since  $\varphi$  has compact support, the boundary terms vanish.

1. For the drift term:

$$\int (f \cdot \nabla \varphi) p_t dx = - \int \varphi \nabla \cdot (f p_t) dx.$$

2. For the diffusion term (applying IBP twice):

$$\int (\Delta \varphi) p_t dx = \int \varphi \Delta p_t dx.$$

Substituting these back yields:

$$\int_{\mathbb{R}^d} \varphi(x) \partial_t p_{t(x)} dx = \int_{\mathbb{R}^d} \varphi(x) \left[ -\nabla \cdot (f(x, t) p_{t(x)}) + \frac{1}{2} g(t)^2 \Delta p_{t(x)} \right] dx.$$

Rearranging terms to one side:

$$\int_{\mathbb{R}^d} \varphi(x) \left[ \partial_t p_{t(x)} + \nabla \cdot (f p_t) - \frac{1}{2} g(t)^2 \Delta p_t \right] dx = 0.$$

Since this equality holds for **any** smooth test function  $\varphi$  with compact support, the fundamental lemma of calculus of variations implies that the term in the brackets must be zero almost everywhere. Thus,  $p_t$  satisfies the Fokker-Planck equation. ■

**Definition 0.1** (Score of the marginal): Define the **score** function as the gradient of the log-density:

$$s_t(x) := \nabla_x \log p_t(x) = \frac{\nabla p_{t(x)}}{p_{t(x)}}.$$

**Theorem 0.2** (Reverse-time SDE): Assume  $p_t$  is smooth and strictly positive for  $t \in (0, T]$ . Let  $\bar{w}_t$  be a standard Wiener process under time reversal.

Then the **reverse-time** process that runs from  $t = T$  down to  $t = 0$  is governed by the SDE:

$$dx_t = [f(x_t, t) - g(t)^2 s_t(x_t)] dt + g(t) d\bar{w}_t,$$

integrated backward in time from  $T \rightarrow 0$ .

*Proof:* Consider the general form of the reverse drift  $\tilde{f}(x, t)$ . For a diffusion process with scalar diffusion  $g(t)$ , the relationship between the forward drift  $f$  and the reverse drift  $\tilde{f}$  is given by:

$$\tilde{f}(x, t) = f(x, t) - \frac{1}{p_{t(x)}} \nabla \cdot (D(t) p_{t(x)}),$$

where  $D(t) = g(t)^2 I$  is the diffusion tensor. Substituting  $D(t)$  into the equation:

$$\begin{aligned} \tilde{f}(x, t) &= f(x, t) - \frac{1}{p_{t(x)}} \nabla \cdot (g(t)^2 p_{t(x)}) \\ &= f(x, t) - g(t)^2 \frac{\nabla p_{t(x)}}{p_{t(x)}}. \end{aligned}$$

Using the definition of the score  $s_{t(x)} = \nabla \log p_{t(x)} = \frac{\nabla p_{t(x)}}{p_{t(x)}}$ , we obtain:

$$\tilde{f}(x, t) = f(x, t) - g(t)^2 s_{t(x)}.$$

Thus, the reverse SDE has the modified drift  $f - g^2 s_t$  and the same diffusion coefficient  $g(t)$ . ■

**Theorem 0.3** (Probability flow ODE): Under the same assumptions, consider the deterministic ODE:

$$\frac{dx_t}{dt} = f(x_t, t) - \frac{1}{2} g(t)^2 s_t(x_t),$$

integrated backward in time from  $T \rightarrow 0$ .

Then this **probability flow ODE** has the **same marginal densities** as the forward SDE.

*Proof:* We show that the probability density evolved by this ODE satisfies the same Fokker–Planck equation as the SDE. Recall the Fokker–Planck equation for the SDE from Theorem 1:

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g(t)^2 \Delta p_t.$$

We use the identity  $\Delta p_t = \nabla \cdot (\nabla p_t)$ . By definition of the score,  $\nabla p_t = p_t s_t$ . Therefore, we can rewrite the diffusion term as:

$$\frac{1}{2} g(t)^2 \Delta p_t = \frac{1}{2} g(t)^2 \nabla \cdot (p_t s_t) = \nabla \cdot \left( \frac{1}{2} g(t)^2 s_t p_t \right).$$

Substituting this back into the Fokker–Planck equation:

$$\begin{aligned} \partial_t p_t &= -\nabla \cdot (f p_t) + \nabla \cdot \left( \frac{1}{2} g(t)^2 s_t p_t \right) \\ &= -\nabla \cdot \left[ \left( f - \frac{1}{2} g(t)^2 s_t \right) p_t \right]. \end{aligned}$$

This equation is exactly the **continuity equation** (Liouville equation) describing the evolution of densities under a deterministic flow field  $v_{t(x)}$ :

$$\partial_t p_t + \nabla \cdot (v_t p_t) = 0.$$

Comparing terms, we identify the velocity field:

$$v_{t(x)} = f(x, t) - \frac{1}{2} g(t)^2 s_{t(x)}.$$

Since the ODE  $\frac{dx_t}{dt} = v_{t(x_t)}$  implies the continuity equation for its marginals, and this equation is identical to the SDE's Fokker–Planck equation, the ODE and the SDE share the same marginal densities  $p_t$ . ■